

Compact Low Pass Filter Design

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1 Introduction

A Compact low pass filter (LPF) with a wide stopband is presented in this report. The filter is implemented on a Rogers RO4003 substrate with a thickness of 1.15 mm, and a dielectric loss tangent of 0.0021.

The proposed design achieves a return loss better than 20 dB with a cutoff frequency of 2.4 GHz, making it highly suitable for ISM band applications. A transmission zero is introduced near the cutoff frequency, resulting in a sharp transition from passband to stopband, this is achieved by introduction of lower L-shaped resonators, which makes the filter highly selective for RF front-end filtering applications. Meanwhile, upper-L shaped resonators suppress high-frequency spurious responses.

The overall filter dimensions are 34.5×15 mm, which makes it well suited for space constrained RF systems such as mobile and portable devices.

Furthermore, the group delay exhibits an almost flat response across the passband, indicating that once the signal passes through the filter, it adds minimal distortion, which is crucial for high quality transmission systems.

Please also note that this design successfully converged across three consecutive simulation targets, reinforcing the reliability and robustness of the results.

2 Filter Design and Structure

2.1 Substrate and Material Specifications

The filter is simulated on a Rogers RO4003 substrate, which is widely used in high-frequency microwave circuits due to its low loss and stable dielectric properties. The key material parameters are summarized in Table 1.

Table 1: Substrate and material specifications.

Parameter	Value	Unit
Substrate material	RO4003	—
Substrate thickness	1.15	mm
Dielectric loss tangent	0.0021	—
Filter dimensions	34.5×15	mm
Cutoff frequency (f_c)	2.4	GHz

2.2 3D Model

The 3D HFSS models of the compact low pass filter are shown in Figure 1 and Figure 2. The Model shown in Figure 1 is initial design of low pass filter with E-field demonstration at 2.4GHz. This design structure shows a strong roll off but has spurious at high frequencies. The design structure shown in Figure 2 eliminates the spurious at high frequencies by adding another the L-shape resonator.

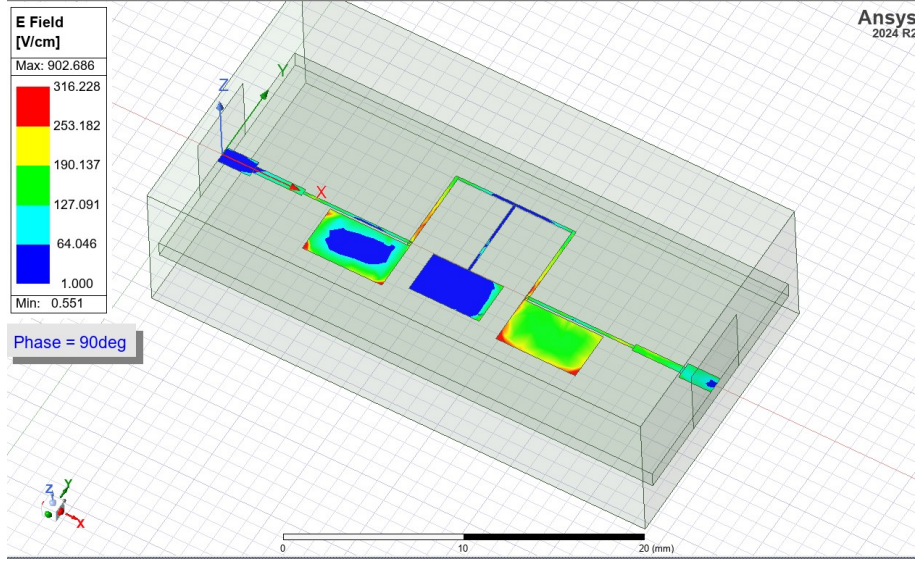


Figure 1: 3D model of the compact low pass filter in Ansys HFSS 2024 R2. Initial Design

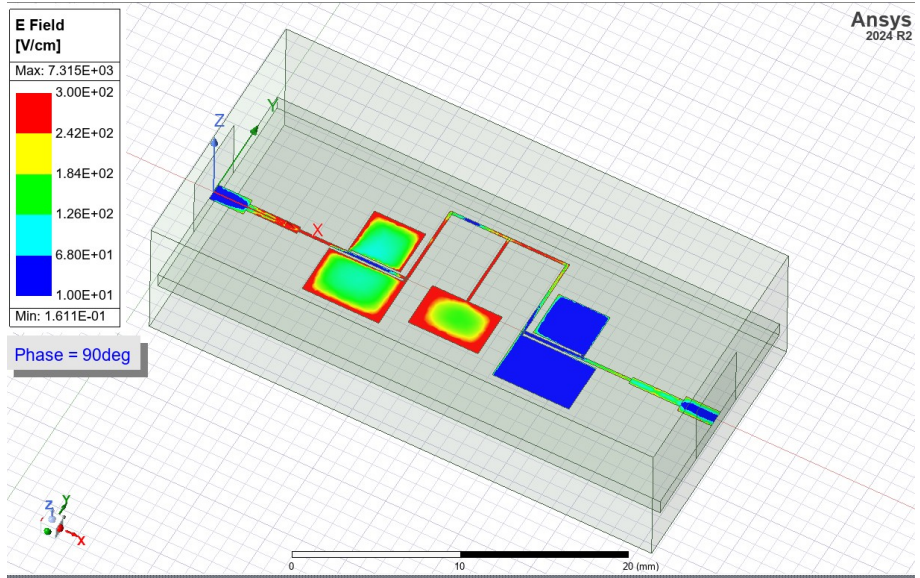


Figure 2: 3D model of the compact low pass filter in Ansys HFSS 2024 R2. Improved Design.

3 Simulation Setup and Convergence

The simulation was carried out using Ansys HFSS 2024 R2 with the Driven Modal solution type. An adaptive meshing strategy was employed. The convergence details are shown in Figure 3.

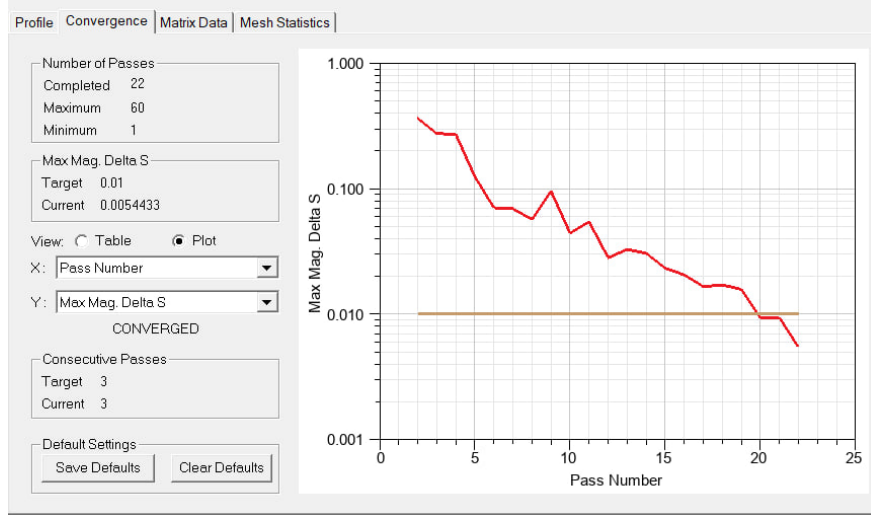


Figure 3: Convergence plot for the adaptive mesh refinement process (compact Low Pass Filter2, Setup1: mahyar1).

The convergence statistics are summarized in Table 2.

Table 2: Adaptive mesh convergence summary.

Parameter	Target	Achieved
Max Mag. Delta S (target)	0.01	0.0054433
Number of passes completed	60 (max)	22
Consecutive passes	3	3 (met)
Convergence status	—	CONVERGED

The simulation converged in 22 adaptive passes, well within the maximum of 60 passes, achieving a Max Mag. Delta S of 0.0054433 (below the target of 0.01) for three consecutive passes. This confirms the numerical reliability of the results.

4 Results and Discussion

4.1 S-Parameter Response of Compact Low Pass Filter-Initial Design

The S-parameter simulation results for the initial design is shown in Figure 4. The orange curve represents $\text{dB}(S_{11})$ (return loss) and the cyan curve represents $\text{dB}(S_{21})$ (insertion loss).

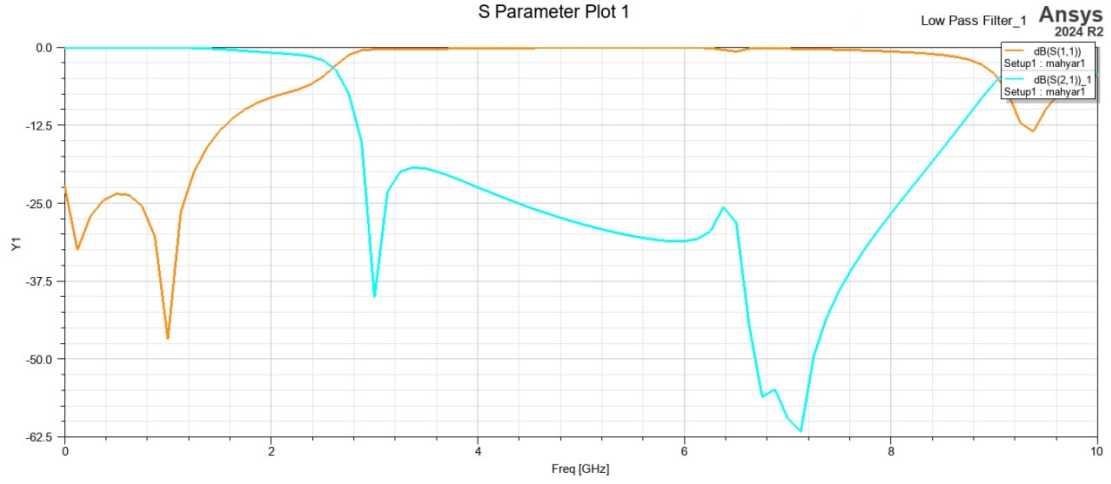


Figure 4: S-parameter plot for Compact Low Pass Filter-Initial Design. Orange: $\text{dB}(S_{11})$; Cyan: $\text{dB}(S_{21})$.

Key observations from Figure 4:

- The -3 dB cutoff frequency is approximately $f_c = 2.40\text{ GHz}$
- The return loss (S_{11}) remains below -20 dB throughout the passband, confirming good input matching.
- A deep transmission null is visible near 2.4 GHz (reaching below -37 dB), creating a sharp transition from passband to stopband.
- Spurious appears between 8 and 10 GHz

4.2 S-Parameter Response of Compact Low Pass Filter-Improved Design

For comparison, the S-parameter response of the improved design is shown in Figure 5. The red curve is $\text{dB}(S_{11})$ and the green curve is $\text{dB}(S_{21})$.

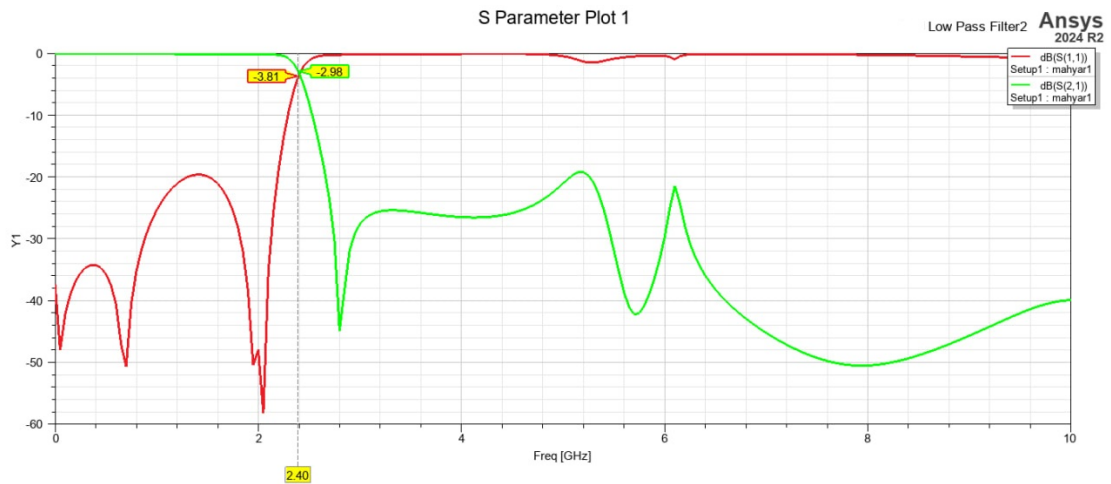


Figure 5: S-parameter plot for Compact Low Pass- Improved Design . Red: $\text{dB}(S_{11})$; Green: $\text{dB}(S_{21})$.

The initial design exhibits a noticeably degraded passband return loss (only ≈ -5 dB near the cutoff) and shows a higher insertion loss across the passband compared to the optimized design and the stopband begins later (around 3 GHz). This comparison highlights the effectiveness of the design optimization. Key observations from Figure 5:

- The -3 dB cutoff frequency is approximately $f_c = 2.40$ GHz (cursor annotation confirms $S_{11} \approx -3.81$ dB, $S_{21} \approx -2.98$ dB).
- The return loss (S_{11}) remains below -20 dB throughout the passband, confirming good input matching.
- A deep transmission null is visible near 2.4 GHz (reaching below -40 dB), creating a sharp transition from passband to stopband.
- The stopband suppression is maintained above -25 dB up to approximately 10 GHz, with minor resonances at around 6 GHz remaining below -20 dB.

4.3 Group Delay

The group delay response of the transmission path (S_{21}) for the optimized filter is shown in Figure 6.

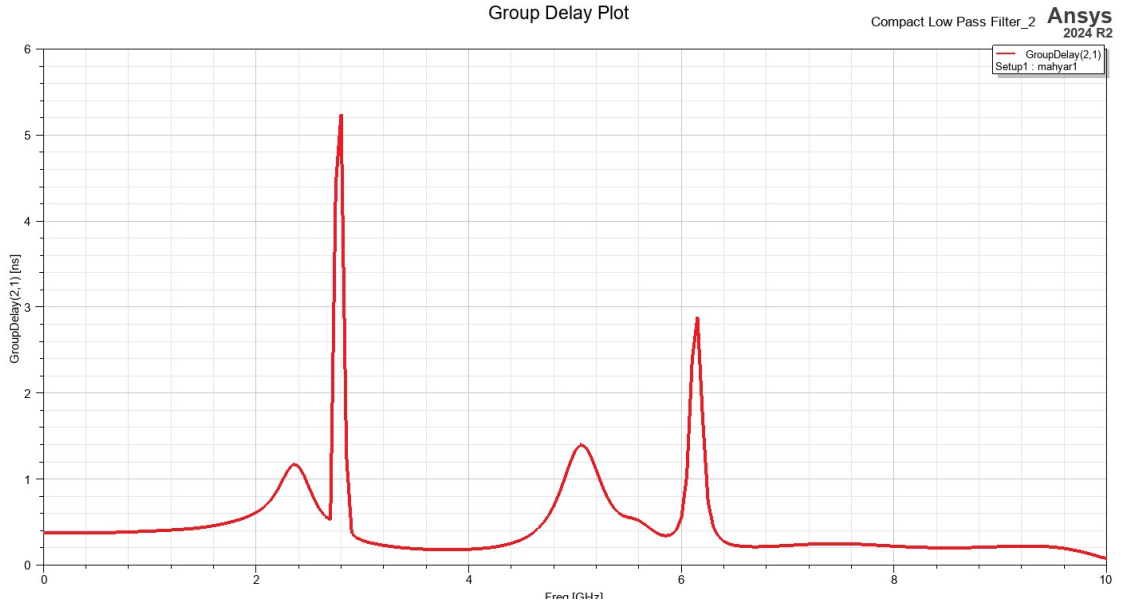


Figure 6: Group delay plot for Compact Low Pass Filter2, dB(GroupDelay(2,1)) vs. frequency.

The group delay is nearly flat in the passband (0–2.4 GHz), with a value of approximately 0.4 ns. This low and stable delay ensures minimal phase distortion for wideband signals and high-quality transmission systems

5 Performance Summary

The key performance metrics of the optimized compact LPF are summarized in Table 3.

Table 3: Performance summary of the compact low pass filter.

Parameter	Value / Result
Substrate	Rogers RO4003
Substrate thickness	1.15 mm
Dielectric loss tangent	0.0021
Filter dimensions	34.5×15 mm
−3 dB cutoff frequency	≈ 2.40 GHz
Return loss in passband	< -20 dB
Transmission null depth	< -30 dB (at ≈ 2.4 GHz)
Stopband suppression (up to 10 GHz)	> 20 dB
Group delay (passband)	Nearly flat (≈ 0.4 ns)
Convergence (Max Mag. Delta S)	0.0054433 (target: 0.01)
Simulation passes	22 of 60 maximum

6 Conclusion

This report presents a compact microwave low pass filter implemented on a Rogers RO4003 substrate with compact dimensions of 34.5×15 mm. The filter achieves a cutoff frequency of 2.4 GHz with a return loss better than 20 dB across the passband, making it highly suitable for ISM band applications and space-constrained systems such as mobile and portable devices.

The use of lower-L shaped resonators creates a sharp transmission null at the passband edge, enabling a fast roll-off into the stopband. Upper-L shaped resonators further suppress high-frequency spurious responses. The group delay is nearly flat in the passband, confirming minimal signal distortion.

The Ansys HFSS adaptive simulation converged robustly after 22 passes (Max Mag. Delta S = 0.0054433), and the design successfully converged across three consecutive simulation targets, reinforcing the reliability and robustness of the results.